D. Ring Road 2

time limit per test: 2 seconds
memory limit per test: 256 megabytes

It is well known that Berland has n cities, which form the Silver ring — cities i and $i+1$ $(1 \le i < n)$ are connected by a road, as well as the cities n and 1. The goverment have decided to build m new roads. The list of the roads to build was prepared. Each road will connect two cities. Each road should be a curve which lies inside or outside the ring. New roads will have no common points with the ring (except the endpoints of the road).

Now the designers of the constructing plan wonder if it is possible to build the roads in such a way that no two roads intersect (note that the roads may intersect at their endpoints). If it is possible to do, which roads should be inside the ring, and which should be outside?

Input

The first line contains two integers n and m $(4 \le n \le 100, 1 \le m \le 100)$. Each of the following m lines contains two integers a_i and b_i $(1 \le a_i, b_i \le n, a_i \ne b_i)$. No two cities will be connected by more than one road in the list. The list will not contain the roads which exist in the Silver ring.

Output

If it is impossible to build the roads in such a way that no two roads intersect, output `Impossible`. Otherwise print m characters. i-th character should be `i`, if the road should be inside the ring, and `o` if the road should be outside the ring. If there are several solutions, output any of them.

Examples

input Copy

```
4 2
1 3
2 4
```

output Copy

```
io
```

input Copy

```
6 3
1 3
3 5
5 1
```

output Copy

```
ooo
```

→ **Attention**

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 27 (Codeforces format, Div. 2)

Finished

Practice

→ **Virtual participation**

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ **Clone Contest to Mashup**

You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ **Last submissions**

Submission	Time	Verdict
372422344	Apr/25/2026 09:49	Accepted
372422033	Apr/25/2026 09:46	Accepted
372421363	Apr/25/2026 09:39	Accepted

372421219	Apr/25/2026 09:38	**Accepted**

→ Problem tags

2-sat | dfs and similar | dsu | graphs | *2200

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1
- Tutorial #2 (en)
- Tutorial #3 (en)

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